

Predictive Lead Scoring System

A Data-Driven Approach to Optimizing Sales Conversion and Efficiency for X Education.

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THE OBJECTIVE



Business Problem

X Education faces high volumes of leads but low conversion efficiency. Manual prioritization is error-prone, leading to wasted sales efforts on "cold" leads.



The Solution

Develop a Logistic Regression model to assign a 'Lead Score' (0-100), enabling the team to focus on high-probability prospects and maximize ROI.

METHODOLOGY PIPELINE



DATA REFINING INSIGHTS

- Placeholder Handling: Converted non-informative "Select" entries into NaNs for accurate missing value analysis.
- Imputation & Removal: Dropped variables with >40% null values. Imputed minor missingness using statistical modes.
- Outlier Management: Capped extreme values in 'Total Visits' and 'Page Views' to ensure model stability.
- Consistency: Trimmed categorical strings to remove hidden whitespaces, ensuring reliable dummy variable creation.



SELECTING DRIVERS



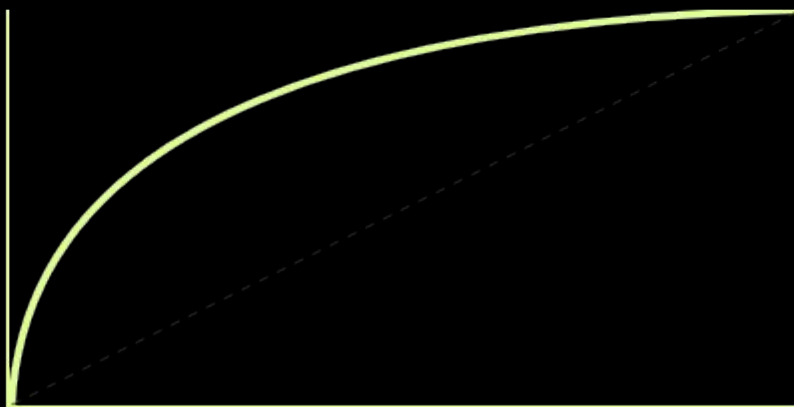
Variables selected based on P-values (<0.05) and VIF (<5) for maximum non-redundant predictive power.

EXCEPTIONAL ACCURACY

0.97

ROC - AREA UNDER CURVE

- **Test Accuracy: 92.84%** – Consistently high performance on unseen lead data.
- **Sensitivity (Recall): 89.99%** – We correctly identify 9 out of 10 potential conversions.
- **Minimal Overfitting:** Gap between Train and Test metrics is <1%, ensuring real-world reliability.



TOP 3 CONVERSION INDICATORS



Horizzon Tag

Leads tagged as "Closed by Horizzon" are the highest value prospects with near-certain conversion.



Email Reverts

Leads stating they "Will revert after reading email" show high active engagement and follow-up intent.



Welingkar Source

The Welingak Website consistently provides superior lead quality compared to organic or generic sources.

PHASE 1: **AGGRESSIVE** HIRING

0.30
SUGGESTED CUTOFF

Strategy: Maximum Coverage

During intern months, we lower the probability threshold to ~0.30. This "casts a wide net," ensuring no potential conversion is missed. The interns provide the bandwidth to process these additional "warm" leads.

PHASE 2: HIGH PRECISION

0.80

SUGGESTED CUTOFF

Strategy: Optimal Efficiency

When targets are met or resources are lean, we raise the threshold to ~0.80. This eliminates "useless" calls, focusing the core sales team exclusively on sure-shot closures while they handle new strategic tasks.

STRATEGIC ACTION PLAN

- **CRM Integration:** Implement the lead score directly into the CRM dashboard for real-time sorting.
- **Automated Alerts:** Setup instant notifications for sales reps when a lead hits a score > 80 .
- **Source Optimization:** Re-allocate marketing budget from generic search to high-performing sources like Welingak.
- **Dynamic Thresholds:** Empower managers to adjust cutoff scores daily based on current intern/staff availability.



KEY ASSUMPTIONS

Assumption 1

Historical behaviors documented in the training set will continue to serve as valid predictors for future lead cohorts.

Assumption 2

Internal processes for "Tagging" leads remain consistent; high-impact variables depend on accurate human input.

Assumption 3

Technical lead sources (Welingak, Landing Page) maintain their current operational quality and user demographics.

Assumption 4

Market conditions remain relatively stable; drastic external shifts may require model re-calibration.

Q&A

Thank you for your time. Are there any questions regarding the Lead Scoring implementation?

Accuracy: 93%

AUC: 0.97

Recall: 90%

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THANK YOU
